Maxwell's equations and energy/momentum of EM field

$$\begin{cases} \vec{\nabla}\cdot\vec{E} = \mu_0 c^2 \rho = \dfrac{\rho}{\varepsilon_0} \\ \vec{\nabla}\times\vec{B} = \mu_0 \vec{j} + \dfrac{1}{c^2}\dfrac{\partial \vec{E}}{\partial t} \\ \vec{\nabla}\times\vec{E} = -\dfrac{\partial \vec{B}}{\partial t} \\ \vec{\nabla}\cdot\vec{B} = 0 \end{cases}$$

$$\vec{\nabla}\times\vec{B}\cdot\vec{E} = \mu_0 \vec{j}\cdot\vec{E} + \frac{1}{c^2}\frac{\partial \vec{E}}{\partial t}\cdot\vec{E}$$

$$-\vec{\nabla}\times\vec{E}\cdot\vec{B} = +\frac{\partial \vec{B}}{\partial t}\cdot\vec{B}$$

$$\Rightarrow \underbrace{\vec{E}\cdot\vec{\nabla}\times\vec{B} - \vec{B}\cdot\vec{\nabla}\times\vec{E}}_{-\vec{\nabla}\cdot(\vec{E}\times\vec{B})} = \mu_0 \vec{j}\cdot\vec{E} + \frac{\partial}{\partial t}\left(\frac{\vec{E}^2}{2c^2} + \frac{\vec{B}^2}{2}\right)$$

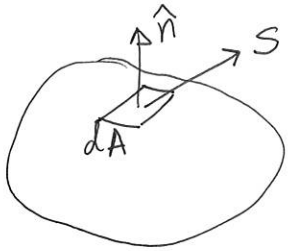
$$\Rightarrow \frac{\partial}{\partial t}\left(\frac{1}{2c^2\mu_0}E^2 + \frac{1}{2\mu_0}B^2\right) + \vec{\nabla}\cdot\left(\frac{\vec{E}\times\vec{B}}{\mu_0}\right) + \vec{j}\cdot\vec{E} = 0$$

$$\frac{1}{c^2} = \varepsilon_0\mu_0$$

$$U = \frac{\varepsilon_0}{2}E^2 + \frac{1}{2\mu_0}B^2 \quad \text{energy density of EM field}$$

$$\vec{S} = \frac{1}{\mu_0}\vec{E}\times\vec{B} \quad \text{Poynting vector}$$

$$\Rightarrow \frac{\partial U}{\partial t} + \vec{\nabla}\cdot\vec{S} + \vec{E}\cdot\vec{j} = 0 \quad \text{local energy conservation}$$



Integrate over volume

$$\frac{d}{dt}\int_V u\,dV + \underbrace{\int dV\,\vec{\nabla}\cdot\vec{S}}_{\oint_A \vec{S}\cdot\hat{n}\,dA} + \int_V \vec{J}\cdot\vec{E}\,dV = 0$$

rate of change of EM energy inside the volume

EM energy flow through the surface enclosing volume

work done on charges inside V by EM field

Conservation of Momentum

$$\vec{B}\times(\vec{\nabla}\times\vec{B}) + \frac{1}{c^2}\vec{E}\times(\vec{\nabla}\times\vec{E}) = \mu_0\,\vec{B}\times\vec{j} + \frac{1}{c^2}\,\vec{B}\times\frac{\partial\vec{E}}{\partial t} - \frac{1}{c^2}\vec{E}\times\frac{\partial\vec{B}}{\partial t}$$

$$\vec{B}\times(\vec{\nabla}\times\vec{B}) + \frac{1}{c^2}\vec{E}\times(\vec{\nabla}\times\vec{E}) + \frac{\partial}{\partial t}\,\frac{1}{c^2}\,\vec{E}\times\vec{B} + \mu_0\,\vec{j}\times\vec{B} = 0$$

Divide by μ_0:

$$\frac{\partial}{\partial t}\left(\varepsilon_0\,\vec{E}\times\vec{B}\right) + \vec{j}\times\vec{B} + \varepsilon_0\,\vec{E}\times(\vec{\nabla}\times\vec{E}) + \frac{1}{\mu_0}\,\vec{B}\times(\vec{\nabla}\times\vec{B}) = 0$$

Subtract $\frac{1}{\mu_0}\,\vec{B}\,(\vec{\nabla}\cdot\vec{B}) = 0$ and $\varepsilon_0\,\vec{E}\left(\vec{\nabla}\cdot\vec{E} - \frac{\rho}{\varepsilon_0}\right) = 0$

$$\Rightarrow \frac{\partial}{\partial t}\left(\varepsilon_0\,\vec{E}\times\vec{B}\right) + \left(\rho\,\vec{E} + \vec{j}\times\vec{B}\right) + \frac{1}{\mu_0}\left[\vec{B}\times(\vec{\nabla}\times\vec{B}) - \vec{B}(\vec{\nabla}\cdot\vec{B})\right]$$

$$+ \varepsilon_0\left[\vec{E}\times(\vec{\nabla}\times\vec{E}) - \vec{E}(\vec{\nabla}\cdot\vec{E})\right] = 0$$

$\Rightarrow$ can identify $\vec{g} = \varepsilon_0\,\vec{E}\times\vec{B}$ with momentum density of EM field

Note $\vec{g} = \varepsilon_0\mu_0\,\vec{S} = \frac{1}{c^2}\,\vec{S}$.

Consider $\left[\vec{B}\times(\vec{\nabla}\times\vec{B}) - \vec{B}(\vec{\nabla}\cdot\vec{B})\right]_i = \varepsilon_{ijk}\,B_j\,(\vec{\nabla}\times\vec{B})_k - B_i\,\frac{\partial B_j}{\partial x_j}$

$$= \varepsilon_{ijk}\,B_j\,\varepsilon_{klm}\,\frac{\partial B_m}{\partial x_l} - B_i\,\frac{\partial B_j}{\partial x_j} =$$

$$\varepsilon_{ijk}\,\varepsilon_{klm} = \varepsilon_{kij}\,\varepsilon_{klm} = \delta_{il}\,\delta_{jm} - \delta_{im}\,\delta_{jl}$$

$$= B_j\,\frac{\partial B_j}{\partial x_i} - B_j\,\frac{\partial B_i}{\partial x_j} - B_i\,\frac{\partial B_j}{\partial x_j} = -\frac{\partial}{\partial x_j}\left(B_i B_j - \frac{1}{2}B^2\delta_{ij}\right)$$

Similarly $\vec{E}\times(\vec{\nabla}\times\vec{E}) - \vec{E}(\vec{\nabla}\cdot\vec{E}) = -\frac{\partial}{\partial x_j}\left(E_i E_j - \frac{1}{2}E^2\delta_{ij}\right)$

$\Rightarrow$ obtain $$\boxed{\frac{\partial g_i}{\partial t} + F_i^{(L)} + \sum_{j=1}^{3}\frac{\partial T_{ij}^{(M)}}{\partial x_j} = 0}$$

$$\vec{F}^{(L)} = \rho\vec{E} + \vec{j}\times\vec{B}$$

$$T_{ij}^{(M)} = -\varepsilon_0 E_i E_j + \frac{\varepsilon_0}{2}E^2\delta_{ij} - \frac{1}{\mu_0}B_i B_j + \frac{1}{2\mu_0}B^2\delta_{ij} =$$

$$= -\varepsilon_0 E_i E_j - \frac{1}{\mu_0}B_i B_j + u\,\delta_{ij}$$

Maxwell stress <u>tensor.</u>

Integrate over volume V: $\quad (d\vec{A} = \hat{n}\,dA)$

$$\underbrace{\frac{d}{dt}\int_V \vec{g}_i\,dV}_{\text{rate of change of EM momentum}} + \underbrace{\int_V \left(\rho\vec{E} + \vec{j}\times\vec{B}\right)_i dV}_{\text{rate of change of particle's momentum inside } V} = -\sum_{j=1}^{3}\int \frac{\partial}{\partial x_j}T_{ij}^{(M)}\,dV =$$

$$= \underbrace{-\sum_{j=1}^{3}\int_A dA\; T_{ij}^{(M)}\hat{n}_j}_{\text{flow of EM momentum through the surface } A \text{ bounding volume } V.}$$

(since $\vec{F}^{(L)} = \frac{d\vec{p}_{part}}{dt}$)

$$-\int dA\sum_{j=1}^{3}T_{ij}^{(M)}n_j = -\int dA\,u\,\hat{n}_i + \int dA\,\varepsilon_0 E_i(E_j n_j)$$

$$+\frac{1}{\mu_0}\int dA\,B_i(B_j n_j) =$$

$$= \left(-\int dA\,u\,\hat{n}\right)_i + \varepsilon_0\int dA\,\vec{E}_i(\vec{E}\cdot\hat{n}) + \frac{1}{\mu_0}\int dA\,\vec{B}_i(\vec{B}\cdot\hat{n})$$

Or in vector notations

pressure $P = \mathcal{U}$ inward

$$\frac{d}{dt}\int \vec{g}\, dV + \int (\rho\vec{E} + \vec{j}\times\vec{B})\, dV = -\int \mathcal{U}\hat{n}\, dA + \varepsilon_0 \int \vec{E}(\vec{E}\cdot\hat{n})\, dA$$

$$+\frac{1}{\mu_0}\int \vec{B}(\vec{B}\cdot\hat{n})\, dA$$

<u>Static field $\vec{E}$:</u>

$$-\mathcal{U}\hat{n} = -\frac{1}{2}\varepsilon_0 E^2 \hat{n}$$

$\varepsilon_0 \vec{E}(\vec{E}\cdot\hat{n})$ = force along $\vec{E}$ with magnitude $\varepsilon_0 \vec{E}\cdot\hat{n}$: max for $\vec{E} \parallel \hat{n}$, 0 for $\vec{E} \perp \hat{n}$.

Note that if $\vec{E} \parallel \hat{n}$ the total electrostatic pressure is $\left(-\frac{1}{2}\varepsilon_0 E^2 + \varepsilon_0 E^2\right)\hat{n}$ along normal to the surface.

Next task is to formulate this in terms of 4-tensors: energy-momentum tensor.